CAB POOLING APPLICATION

A Desktop Application for Efficient Travel Management

CS1201-Mini Project
Priyadeep Jaiswal
2401EC10

Problem Description

- **Current Challenges:**

  High transportation costs for individuals

  Environmental impact of single-occupancy vehicles

  Difficulty in finding travel companions

  Inefficient use of vehicle capacity

- **Impact:**

  Increased travel expenses

  Environmental pollution

  Traffic congestion

  Wasted resources

Proposed Solution

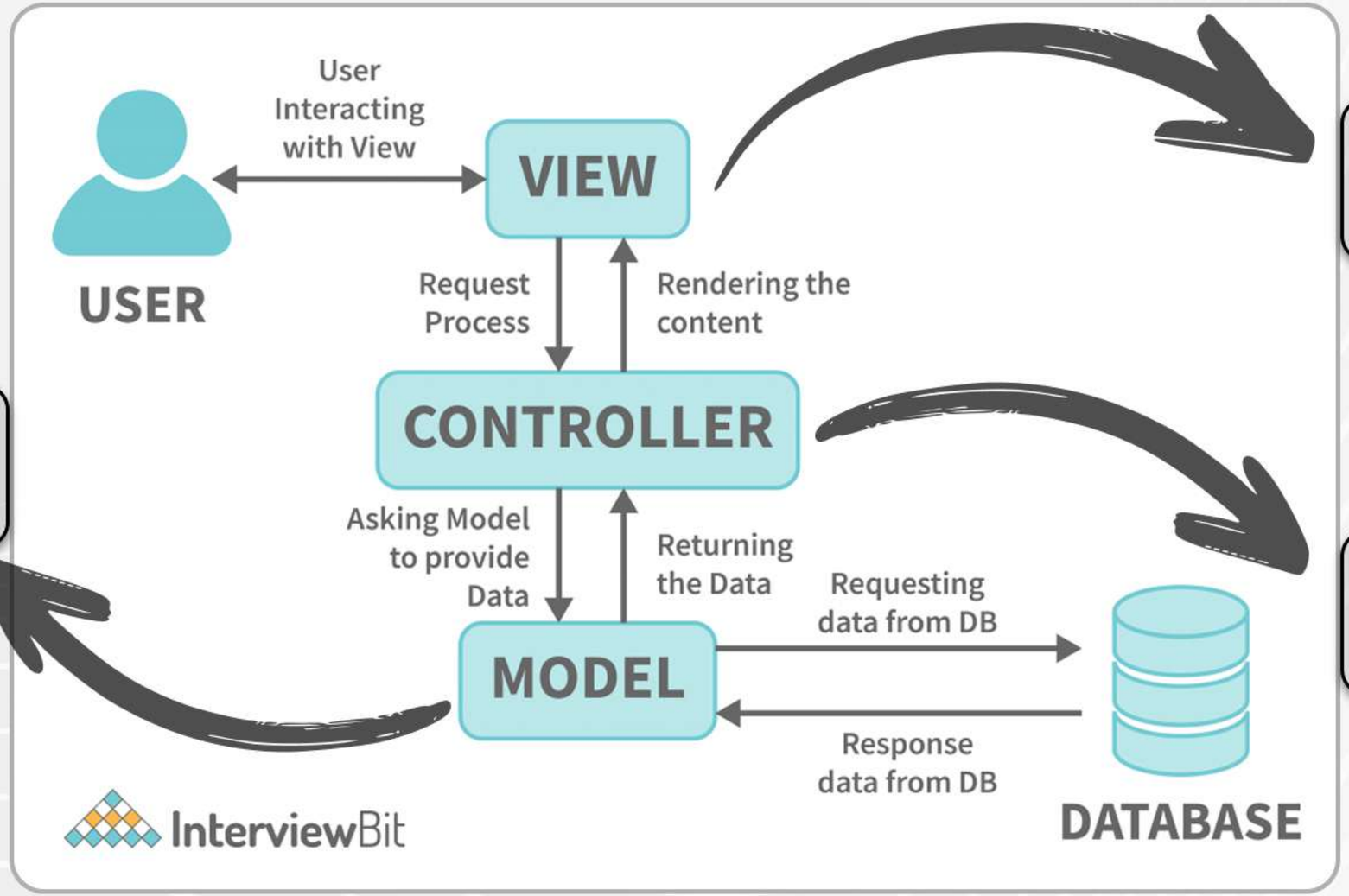
Features

- User-friendly interface
- Secure user authentication
- Efficient search functionality
- Contact information sharing

Benefits

- Cost reduction
- Environmental sustainability
- Community building
- Resource optimizaition

System Architecture



View:
- PyQt5 GUI
- User Interface

Model:
- Database (SQLite)
- Data Classes

Controller:
- Business Logic
- Data Management

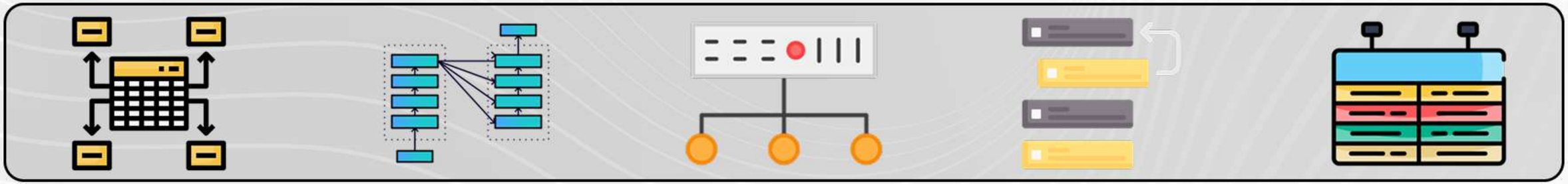
Data Structures Implementation

1. Database Tables:
- Users
- Trips
- Trip Participants

2. Python Classes:
- User
- Trip

3. GUI Components:
- Widgets
- Layouts
- Event Handlers

4. Data Management:
- Lists
- Dictionaries
- Trees (Database Indexing)



Key Features

👤 User Management

- Registration
- Login
- Profile Management

🚗 Trip Management

- Create Trips
- Search Trips
- Join Trips
- Delete Trips

🔍 Search Functionality

- Location-based
- Date-based
- Vehicle type
- Available seats

🔔 Real-time Updates

- Trip status
- Seat availability
- Participant information

Technical Implementation

💾⚙️🛠️ Backend

- Python 3.x
- SQLite Database
- MVC Architecture

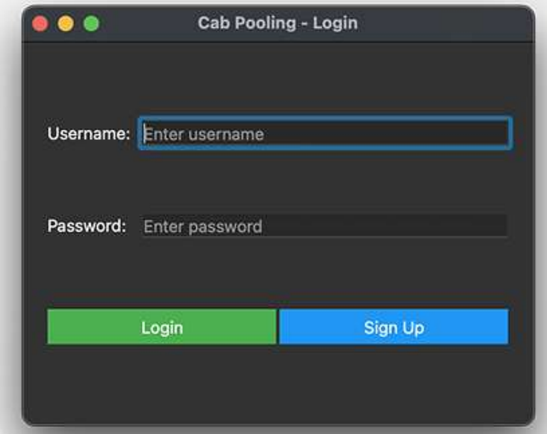
🖥️🎨🧩 Frontend

- PyQt5 Framework
- Custom Widgets
- Responsive Design

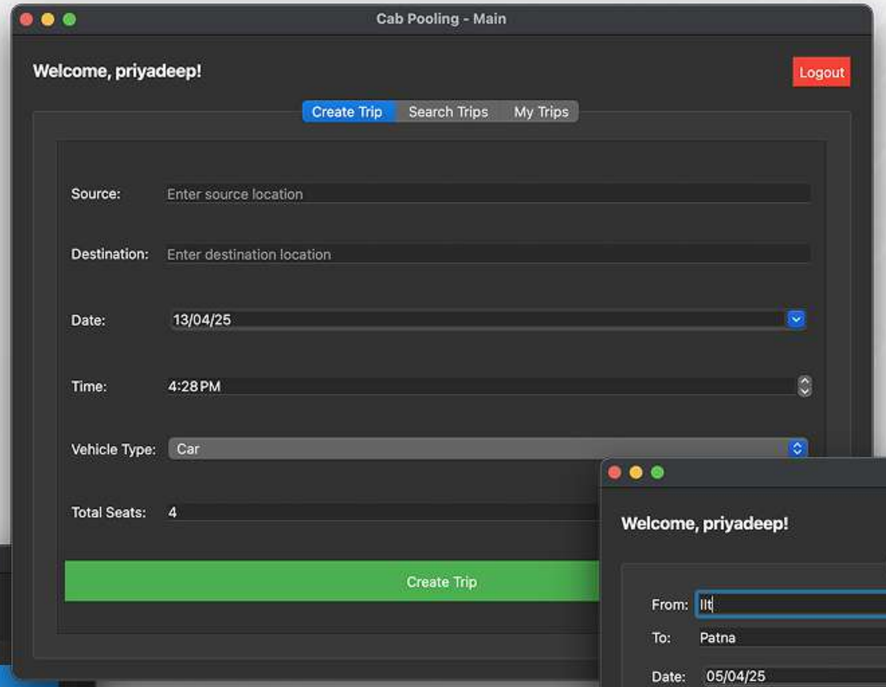
🔐👁️👨‍💻 Security

- Password Hashing
- User Authentication
- Data Validation

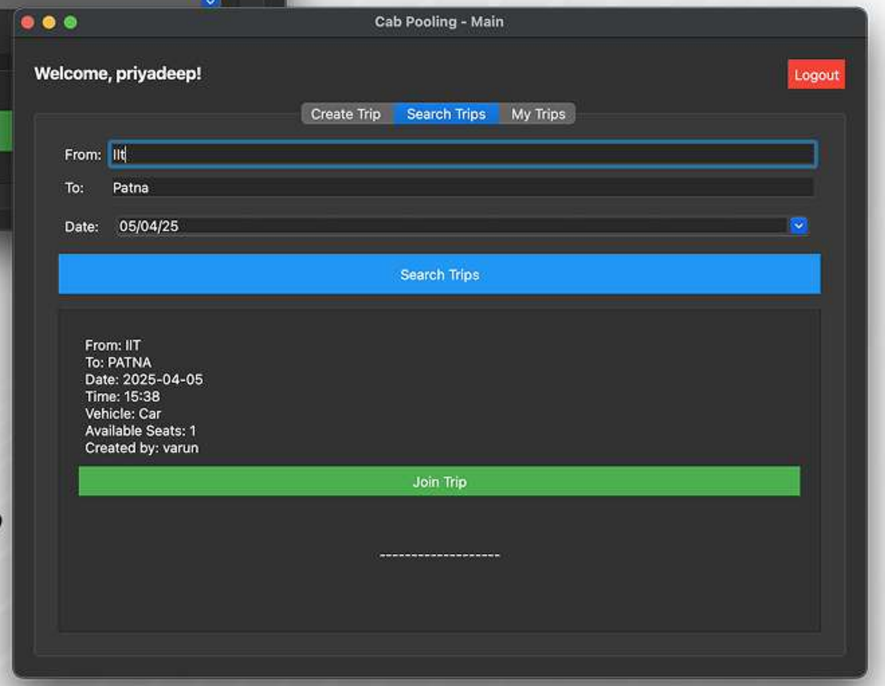
User Interface



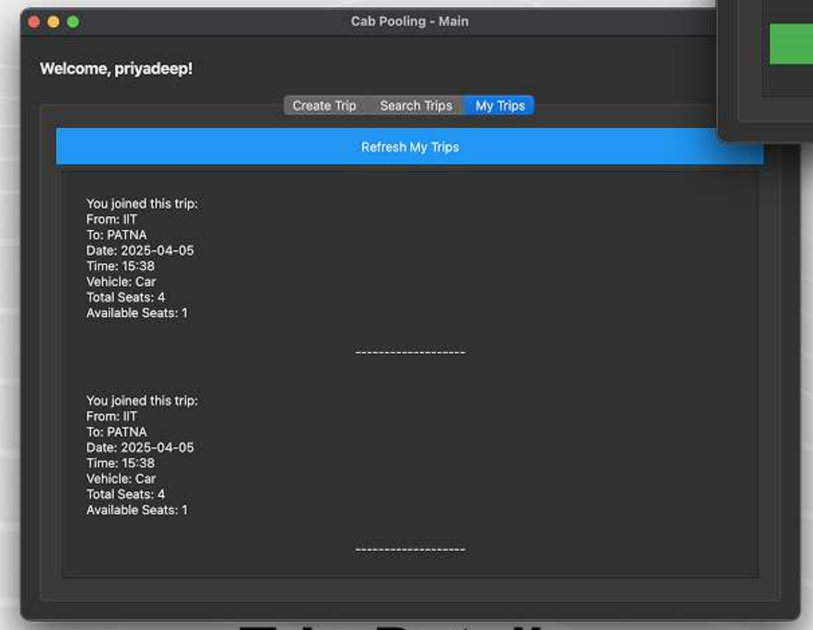
Login/Registration



Trip Creation

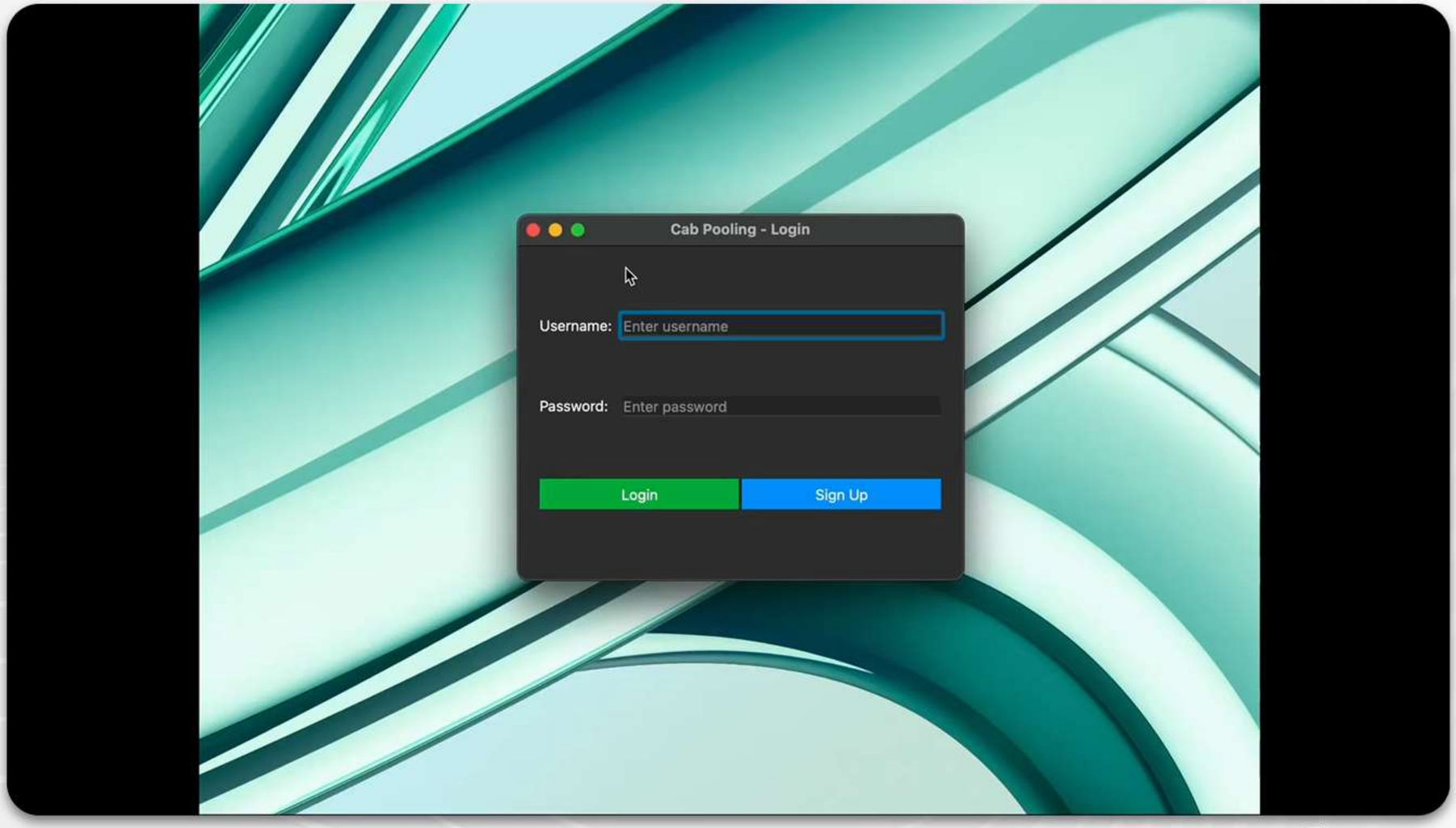


Trip Search



Trip Details

Live Demonstration



Future Scope

Planned Improvements:

1. *Mobile Application*
2. *Payment Integration*
3. *Rating System*
4. *Chat Functionality*
5. *Route Optimizaition*
6. *Real-time Location Tracking*